No.

Date.

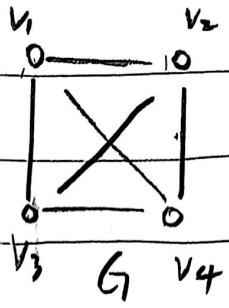
Homework #8

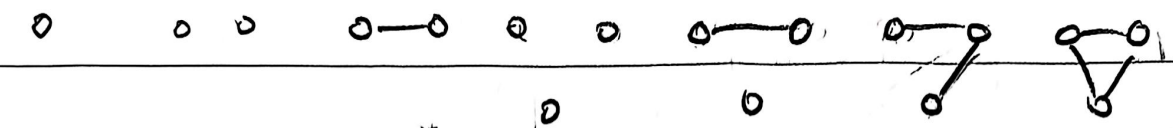
T4. 6, 15, 16, 19, 31, 33, 37, 45

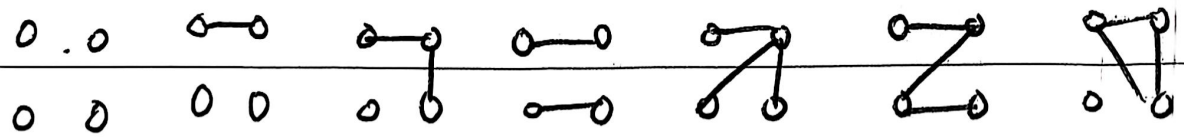
T4 (1) $1\times2^6+4\times2^3+6\times2+4+1=121$,

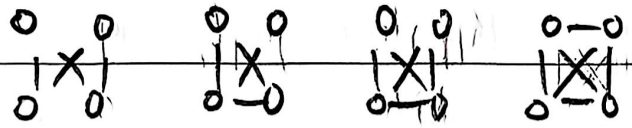
(2) $2^6=64$

(3) $1+2+4+11=18$.

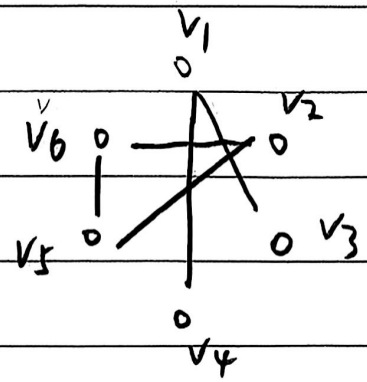








T6

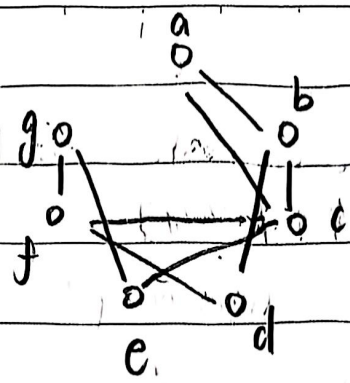


G是非连通的

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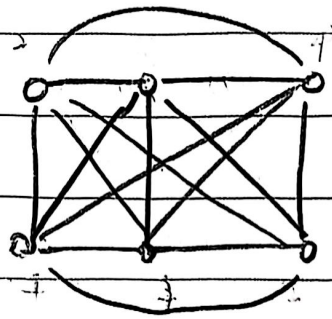
T15



此图连通

故任意两人都能交谈

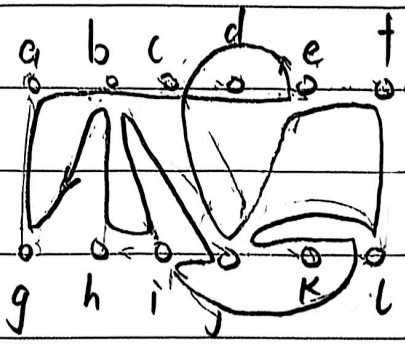
T19



图G的闭包为完全图

故图G为哈密顿图

T16



即 abcdecjefljKlij

bihbga

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T31 证明：若图G不连通，令G=(V,E)

则 $V=\bigcup_{i=1}^{k} V_i$，其中，V_i与V_j之间无边相连($i\neq j$，$k\geq 2$)

任取两结点u,v，若u,v∈V_i，取w∈V_j($i\neq j$)

由于V_i,V_j间无边相连，故在补图$\overline{G}$中有(u,w),(v,w)，即u,v连接

若$u\in V_i$，$v\in V_j$，由于图G中u,v不连接

故在补图$\overline{G}$中u,v连接

故补图$\overline{G}$中任意两结点连接，补图$\overline{G}$是连通的

T33. $$S=\sum_{i=1}^{n} deg(V_i)=2m=2n-2$$

假设T中叶结点数为p，则 $S\geq p+2(n-p-1)+k$

则有 $\Rightarrow p\geq k$

即T中至少有k个叶结点

T37. 先根：$v_0v_1v_4v_5v_2v_3v_6v_8v_9v_{10}v_7$

中根：$v_4v_5v_1v_2v_8v_9v_{10}v_6v_7v_3v_0$

后根：$v_5v_4v_{10}v_9v_8v_7v_6v_3v_2v_1v_0$

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T45 若G中任意结点v, $deg(v) \geq 6$

则由Lemma $S = \sum_{i=1}^{n} deg(v_i) \geq 6n$

即 $m \geq 3n$, 这与 $m \leq 3n-6$ 矛盾

故G中必有一结点v, $deg(v) \leq 5$